Parul® University
Vadodara, Gujarat
NAAC GRADE A++

Information and
Communication Technology

Introduction to Cloud Computing and AWS

Assi. Prof. Jayshree Ranoliya
Assistant Professor
Artificial Intelligence and Data Science Department

Parul® University
Vadodara, Gujarat

NAAC GRADE A++

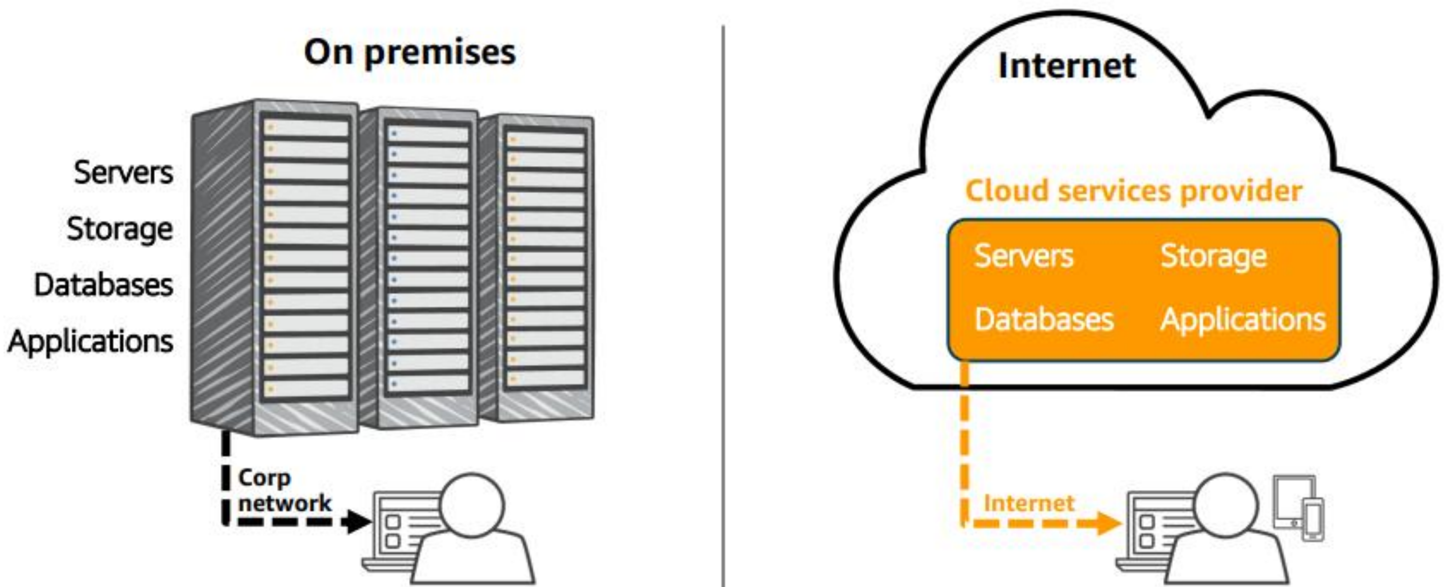
Information and Communication Technology

Content

1. What is Cloud Computing……………………………………………….03
2. Key Benefits of Cloud Computing………………………………….06
3. Cloud Service Models…………………………………………………….12
4 . Cloud Deployment Models…………………………………………….15
5. What is AWS?……………………………………………………………….31
6. The AWS Global Infrastructure……………………………………..32

What is Cloud Computing?

- Cloud computing is a technology that allows users to access computing resources such as servers, storage, databases, networking, software, and analytics over the internet instead of using local computers or on-premise infrastructure.



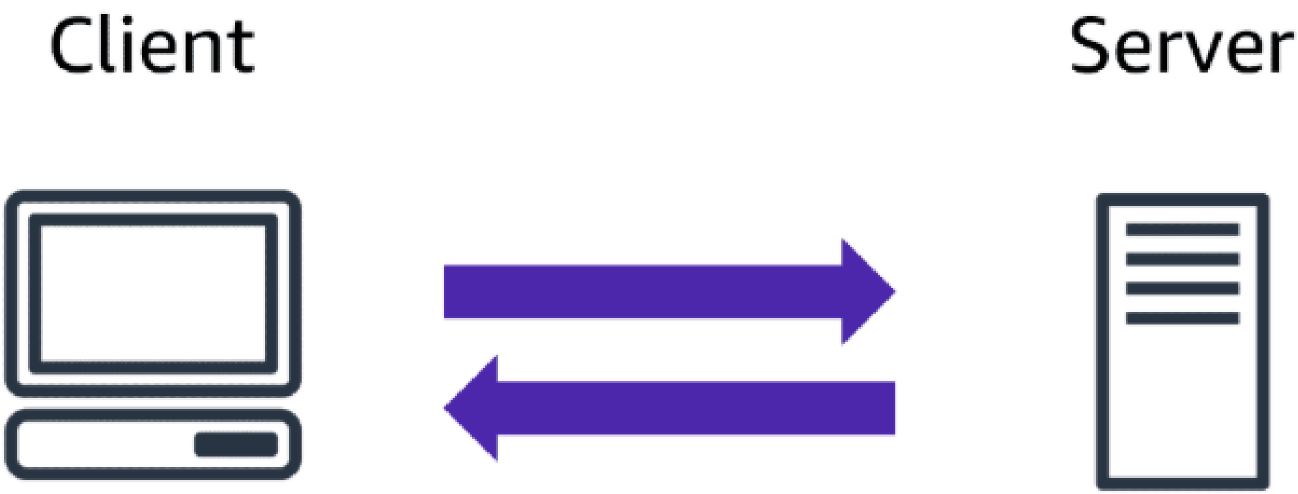
What is Cloud Computing?

- Cloud services are typically offered on a **pay-as-you-go model**, meaning users only pay for the resources they consume.
- Provides development platforms, storage, software, and databases online.
- Supports multiple service models and deployment models.
- Offers scalability, flexibility, security, and reduced costs.
- Eliminates the need for direct infrastructure management.

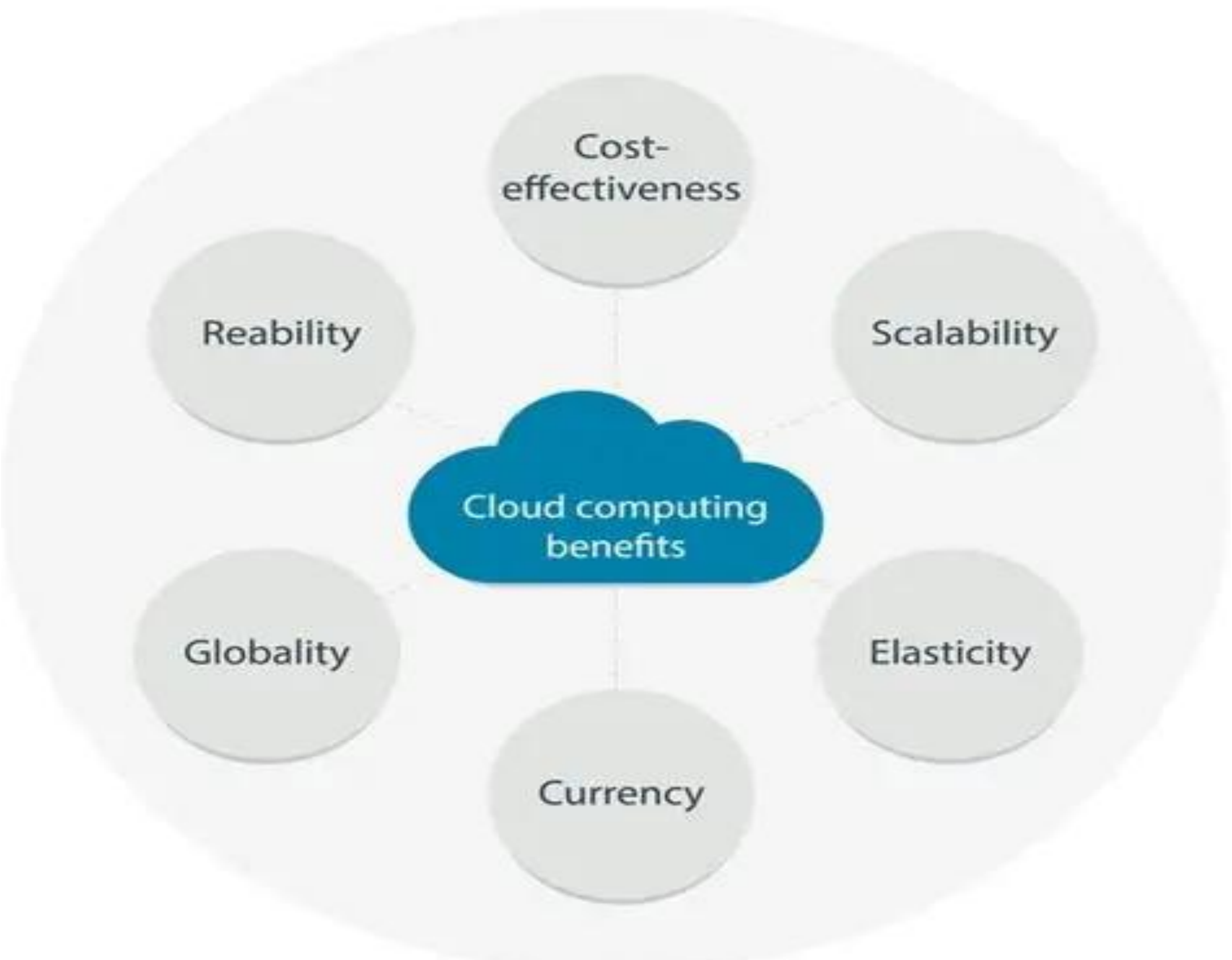
What is Cloud Computing

Cloud computing and the Client-Server Model

Cloud computing is built on the client-server model. Your device (the client) sends requests over the internet to remote servers, which process them and return results.



Key Benefits of Cloud Computing



Key Benefits of Cloud Computing

Cost Efficiency

- Eliminates the need for large upfront investments in hardware.
- Pay-as-you-go pricing ensures you only pay for what you use.
- Reduces maintenance and operational costs.

Scalability and Flexibility

- Instantly scale resources up or down based on demand.
- Supports dynamic workloads and seasonal traffic spikes.
- Enables global expansion without physical infrastructure.

Key Benefits of Cloud Computing

Security and Reliability

- Advanced encryption, firewalls, and compliance standards.
- Regular updates and monitoring ensure data protection.
- Redundant systems minimize downtime and data loss.

Accessibility and Collaboration

- Access data and applications from anywhere, anytime.
- Facilitates teamwork through shared resources and tools.
- Enhances productivity with real-time collaboration.

Key Benefits of Cloud Computing

Automatic Updates and Maintenance

- Providers handle software updates and infrastructure maintenance.
- Ensures systems stay current and secure without user intervention.

Innovation and Speed

- Rapid deployment of applications and services.
- Encourages experimentation with minimal risk.
- Accelerates time-to-market for new solutions.

Cloud Service Models

- In this section we will learn about **cloud computing models** and **cloud computing deployment models**
- Cloud services are generally divided into three major models.

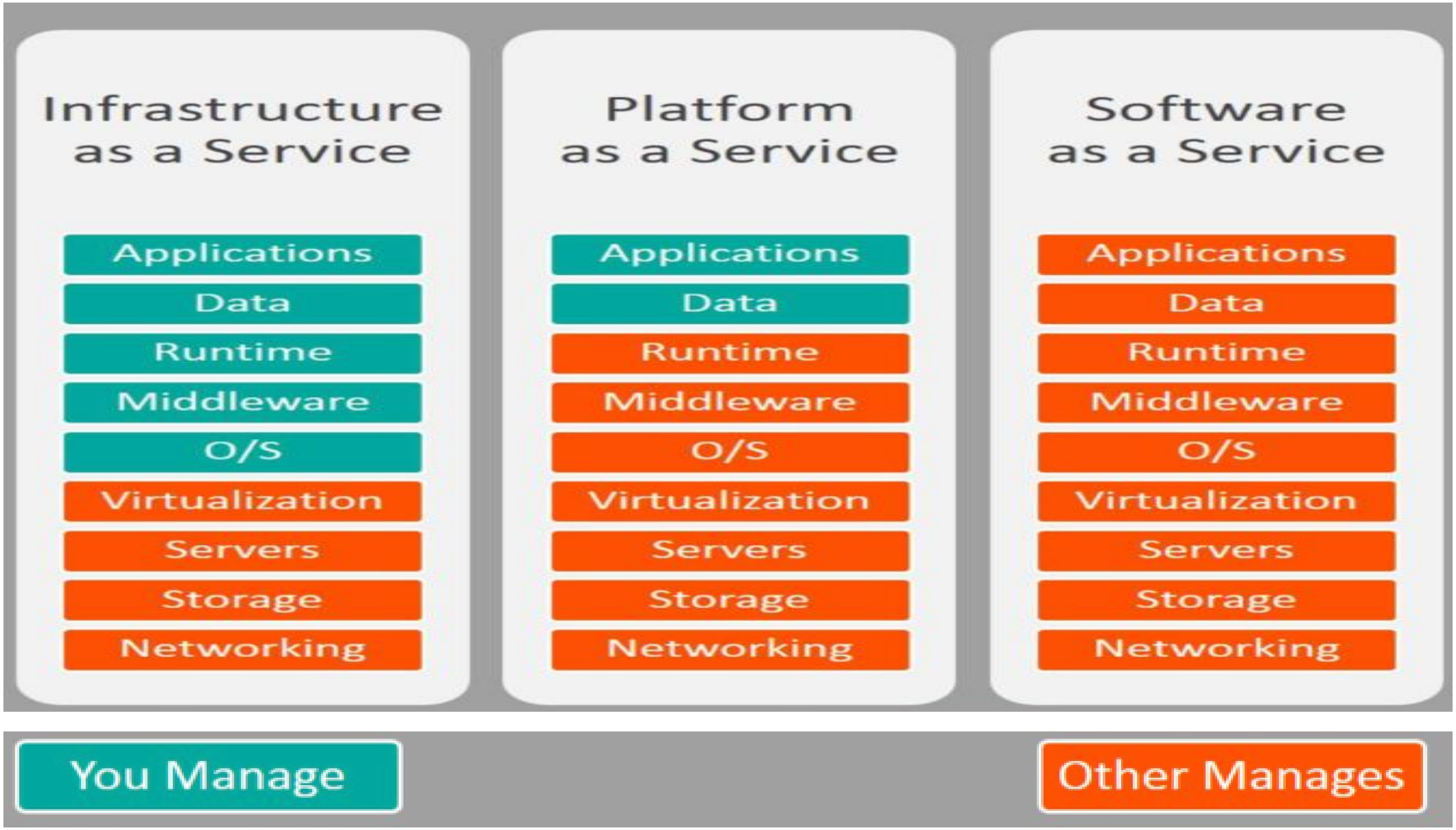
1. **Infrastructure as a Service (IaaS)**
2. **Platform as a Service (PaaS)**
3. **Software as a Service (SaaS)**

Cloud Service Models

- In this section we will learn about **cloud computing models** and **cloud computing deployment models**
- Cloud services are generally divided into three major models.

1. **Infrastructure as a Service (IaaS)**
2. **Platform as a Service (PaaS)**
3. **Software as a Service (SaaS)**

Cloud Service Models



Software as a Service (SaaS): It's what you *use* when you choose to use the cloud. For example — Gmail. All you do is manage the inbox, and Google takes care of the data centers, servers, networks, storage, maintenance, patching, etc. all you worry about is the software itself and how you want to use it. I couldn't find an example for an AWS service that is a SaaS, so I assume there isn't (yet).

Platform as a Service (PaaS): It's what you *build on top of* when you use the cloud. Someone else is managing the underlying hardware and operating system (security patching, update, maintenance). You just focus on your applications.

Real-life example: Heroku! It’s a platform where you upload your project and they store and run it. You receive a URL (or buy your own and attach) and you have your website.

An AWS example for PaaS is elastic beanstalk. We will meet that service in future blog posts.

Infrastructure as a Service (IaaS): It’s what you *migrate to* when you choose to use the cloud. You manage the server which can be physical or virtual and the data center provider will have no access to your server.

A real-life example is a hospital building: the floors, the rooms, the elevator… these are not medical but essential for the operations.

An AWS example for IaaS is EC2. It’s a virtual server and we will talk about it soon.

Cloud Deployment Models

A cloud deployment model explains where your data, apps, and services run and who has control over them. It helps decide how secure, flexible, and cost-effective your setup can be. **There are four main cloud deployment models**.

1. **Public cloud:** A provider owns and manages the resources, and anyone can use them.
2. **Private cloud:** Only one company uses the infrastructure, so it offers better control and security.
3. **Hybrid cloud:** It mixes public and private clouds, allowing movement of data between the two.
4. **Multi-cloud:** It uses multiple cloud providers to avoid depending on just one

Cloud Deployment Models

1. Public Cloud

The public cloud model is a type of cloud deployment model where services are offered by third-party providers over the internet. It allows businesses to access computing resources without managing physical infrastructure. This model is flexible, cost-effective, and suited for many use cases.

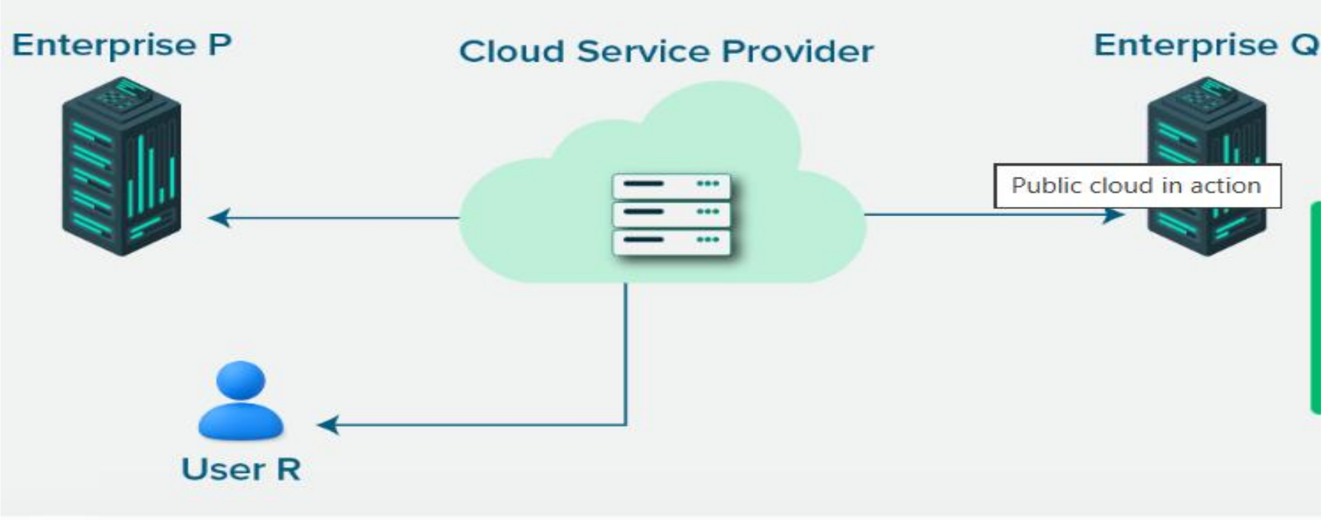


Fig : Public Cloud Deployment Model

Public Cloud Deployment Model

Commonly used public clouds include Microsoft Azure, Amazon AWS, Google Cloud, Oracle Cloud, and Alibaba Cloud.

Advantages

Scalability: Resources can be scaled up or down instantly to match workload demands

Cost efficiency: Pay-as-you-go pricing reduces upfront capital expenses and aligns costs with actual usage

Reduced management overhead: The provider handles maintenance, updates, and infrastructure reliability

Rapid deployment: New environments and applications can be provisioned in minutes.

Public Cloud Deployment Model

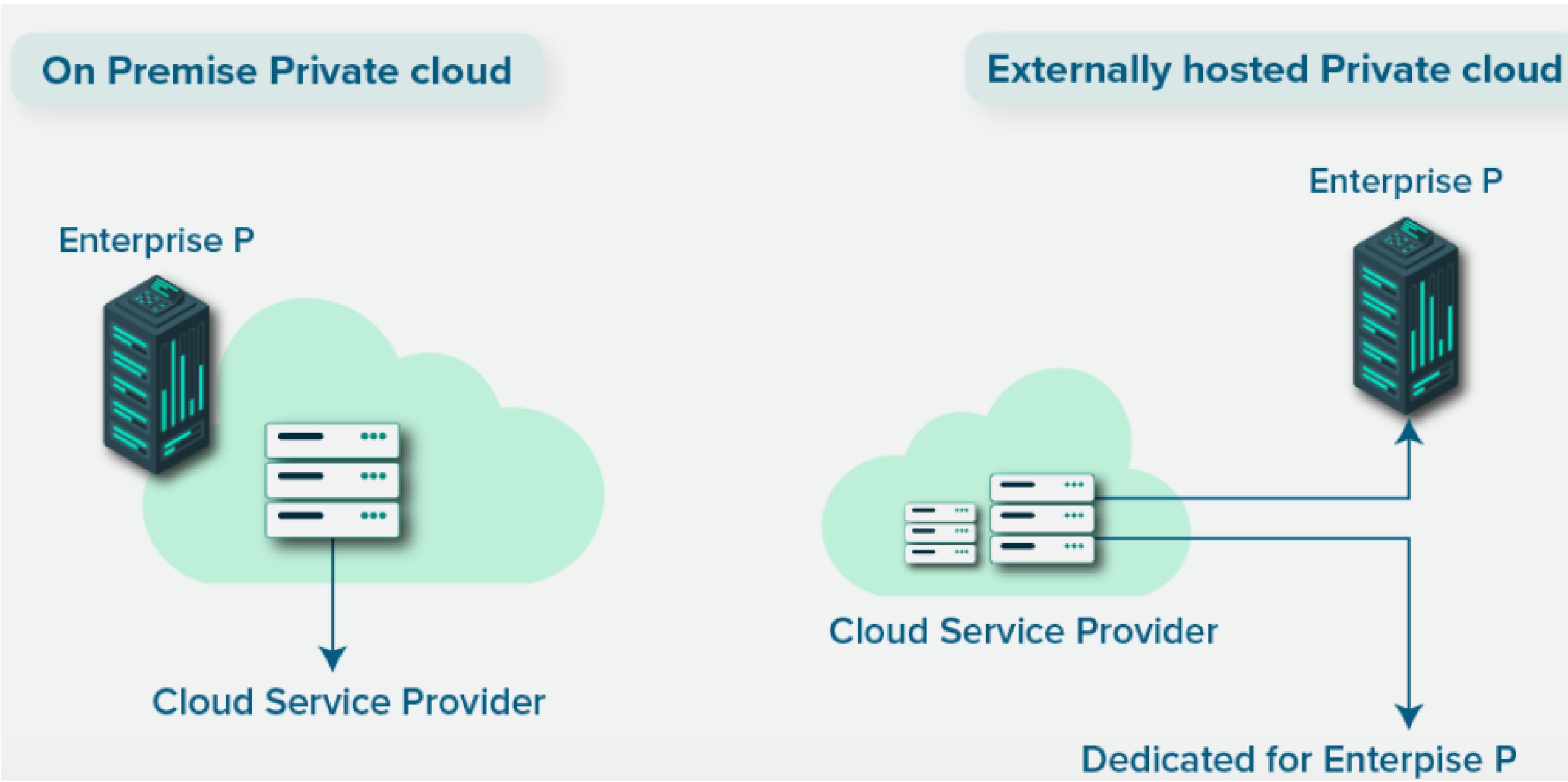
Global availability: Data centers across regions improve performance and support international operations

Flexibility: Ideal for variable workloads, testing environments, and disaster recovery solutions

Disadvantages of the Public Cloud Model

- **Security concerns**: Shared infrastructure increases exposure to risks, requiring strict compliance and data protection measures
- **Limited control**: Organizations have less visibility and customization compared to private or on-premises environments
- **Unpredictable costs**: Usage-based billing can lead to unexpectedly high expenses if workloads are not carefully monitored

Private Cloud Deployment Model



Private Cloud Deployment Model

The private cloud is a type of cloud deployment model where computing resources are dedicated to a single organization. These resources can be hosted on-premise or by a third party. Unlike the public cloud model, the private cloud model offers greater control, privacy, and customization, making it suitable for companies with strict security or compliance needs.

Advantages of the private cloud model.

Enhanced security: Dedicated infrastructure reduces risks associated with multi-tenancy and supports strict compliance

Full control: Organizations manage configuration, resources, and policies to match specific business needs

Private Cloud Deployment Model

Advantages of the private cloud model.

Predictable performance: Resources are not shared with other tenants, ensuring consistent availability and reliability

Customization: Infrastructure and services can be tailored to unique workloads or legacy applications

Regulatory compliance: Easier to meet industry standards for data residency, privacy, and auditing

Private Cloud Deployment Model

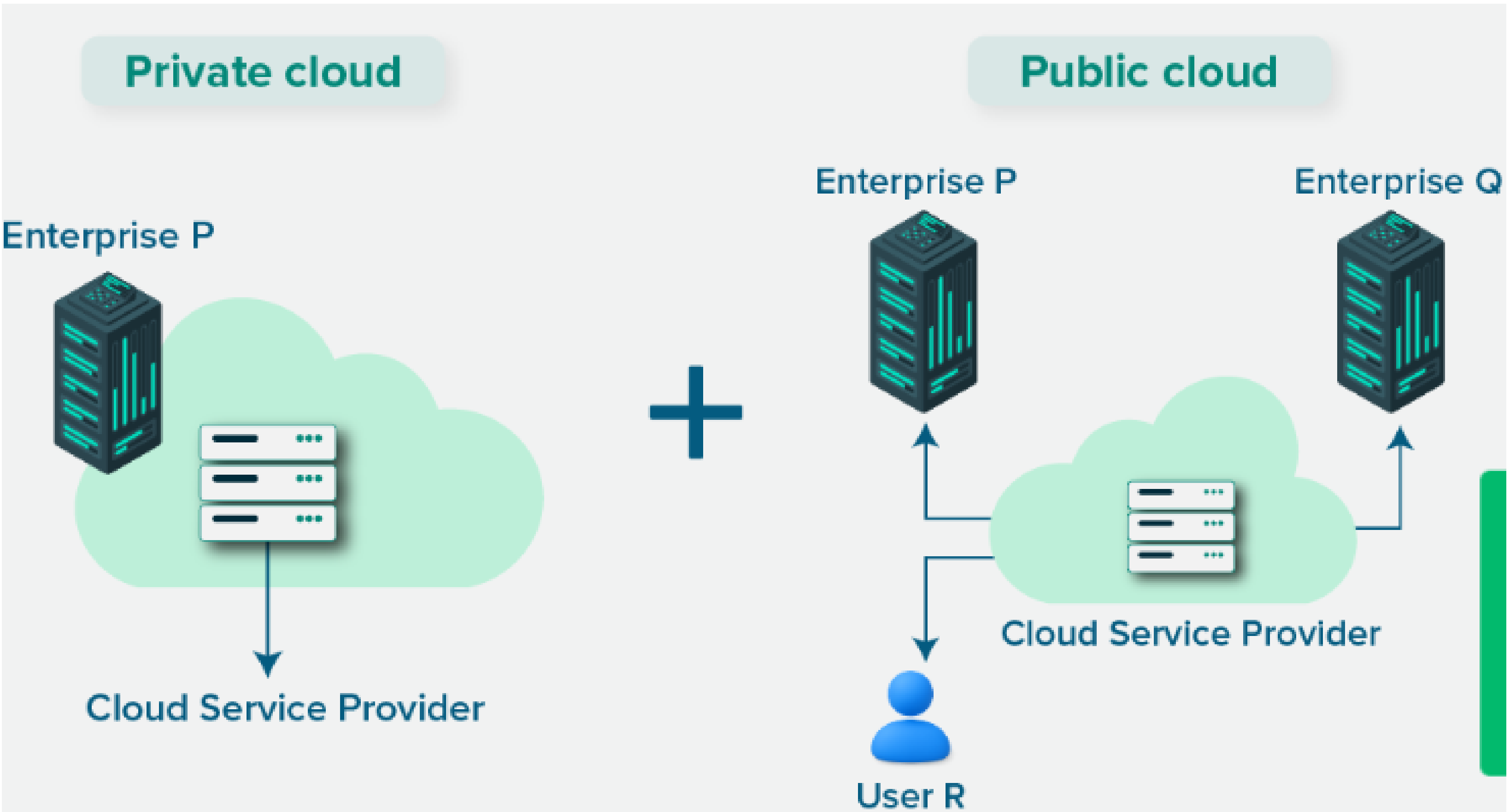
Disadvantages of the private cloud model.

High upfront cost: Significant investment is required for hardware, software, and data center infrastructure

Complex management: Organizations must handle maintenance, upgrades, and security internally, increasing operational burden

Limited scalability: Expanding capacity is slower and more expensive compared to the elasticity of public cloud services

Hybrid Cloud Deployment Model



Hybrid Cloud Deployment Model

The hybrid model **combines both public and private cloud deployment models**, giving a single cloud infrastructure that is aimed at increasing flexibility and deployment options for the business.

For example, applications with strict governance and data security requirements may be hosted in the business’s private cloud, whereas applications without these concerns, which need to be scaled on demand, could be hosted in the public cloud.

Advantages of the Hybrid Cloud

Flexibility: Workloads can move between private and public environments depending on performance, security, or cost needs

Cost optimization: Sensitive data can remain on private infrastructure while variable workloads use the public cloud’s pay-as-you-go model

Scalability: Public cloud resources provide on-demand capacity during peak usage, reducing pressure on private infrastructure

Business continuity: Data and applications can be distributed across environments, improving disaster recovery and resilienc

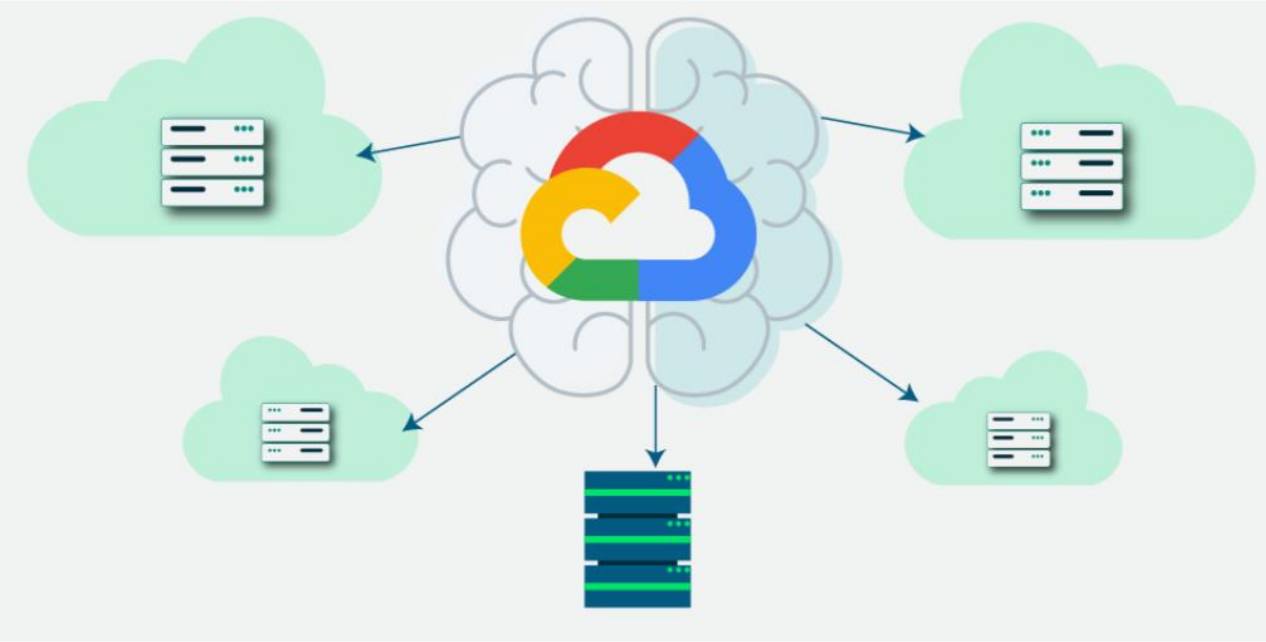
Disadvantages of the Hybrid Cloud

Integration complexity: Managing interoperability between public and private environments requires advanced networking, security, and monitoring.

Higher costs: Maintaining private infrastructure alongside public cloud services can increase overall expenses compared to a single model.

Multi - Cloud Deployment Model

The multi-cloud is a type of cloud deployment model where a business uses services from more than one cloud provider. It combines public or private cloud platforms from different vendors. This model helps avoid reliance on a single provider and offers more control over specific workloads.



Advantages of the Multi- Cloud Model

Vendor independence: Reduces reliance on a single provider, avoiding lock-in and improving negotiation leverage

Resilience and availability: Workloads can be distributed across providers, minimizing downtime from regional or platform-specific failures

Optimized performance: Organizations can choose the best services from different providers to match specific workload requirements

Disadvantages of the Multi- Cloud Model

Management complexity: Coordinating multiple providers requires advanced monitoring, governance, and integration tools

Increased costs: Duplicated services, data transfer fees, and specialized management can drive higher overall expenses

What is AWS?

Amazon Web Services (AWS) is the world's most comprehensive and broadly adopted cloud platform. Since its launch in 2006, AWS has revolutionized how individuals, companies and governments access technology services.

- Access computing power, storage, and databases on-demand
- Eliminate the need to buy, own, and maintain physical data centers
- Pay only for what you use (pay-as-you-go model)
- Flexible scaling based on business needs

AWS Global Infrastructure

AWS provides a reliable, scalable, and low-cost infrastructure platform in the cloud that powers hundreds of thousands of businesses in 190 countries around the world.

Components of AWS Global Infrastructure:

1. **Regions:**
   - Physical locations around the world where AWS clusters data centers.
   - Currently **39 Regions** with more planned.
   - Each Region has multiple Availability Zones.

AWS Global Infrastructure

Components of AWS Global Infrastructure:

1. **Regions:**

   •AWS has Regions all around the world

   •Names can be us-east-1, eu-west-3…

   •A region is a **cluster of data centers**

   •Most AWS services are region-scoped

AWS Global Infrastructure

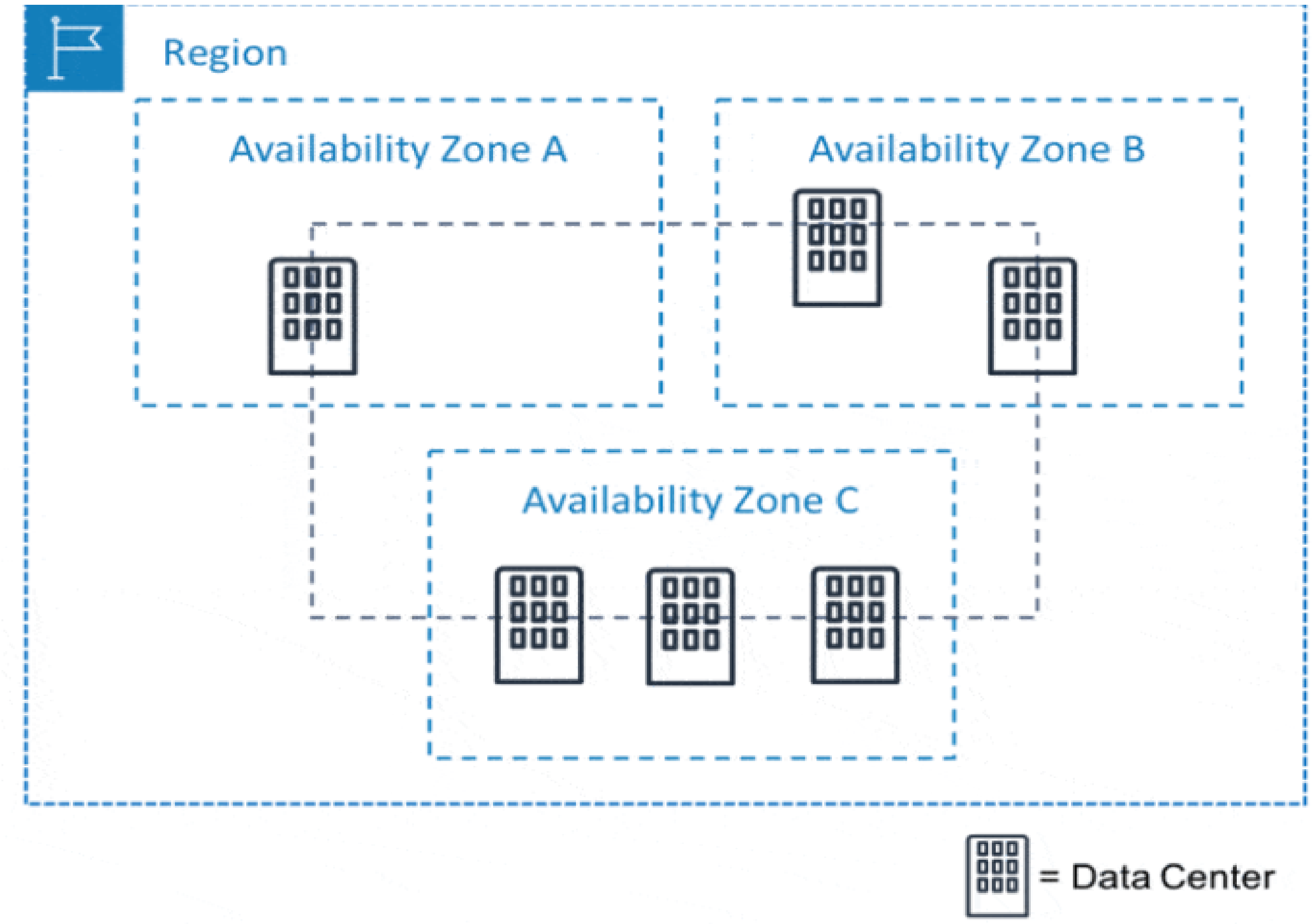
Components of AWS Global Infrastructure:

2. Availability Zones

Within each region, AWS's infrastructure is further fortified through the concept of availability zones. An availability zone represents a discrete data center facility equipped with its own power, cooling, and networking infrastructure. These zones are isolated from each other to ensure fault tolerance and high availability. In the event of hardware failures or disruptions, applications can be seamlessly transitioned to alternate availability zones, thereby safeguarding operational continuity.

Components of AWS Global Infrastructure:

2. Availability Zones



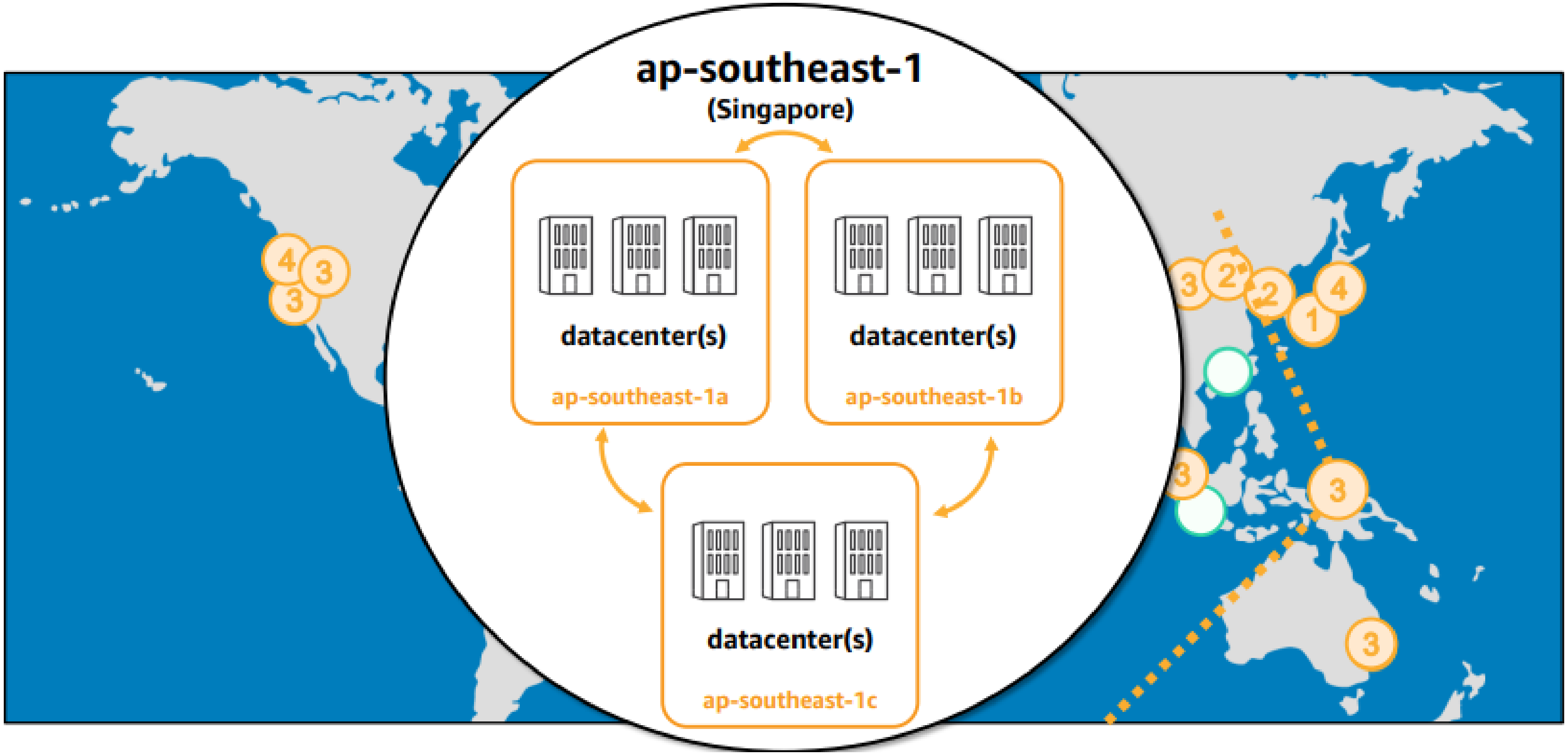
Components of AWS Global Infrastructure:

2. Availability Zones

- When choosing availability zones for your AWS resources, there are a few factors to consider, such as:
- Redundancy: You should choose availability zones that are in different geographic locations to provide redundancy in case of a natural disaster.
- Latency: You should choose availability zones that are close to your users to improve performance.
- Cost: The cost of AWS services varies by availability zone. For example, services in some availability zones are more expensive than services in other availability zones.

Components of AWS Global Infrastructure:

2. Availability Zones



3. Edge Location : Optimizing Content Delivery

To enhance content delivery and minimize latency, AWS employs a network of edge locations strategically positioned in various global cities. *These edge locations serve as points of presence, caching content, and improving the distribution of data to end users.* Amazon CloudFront, AWS's content delivery network, capitalizes on these edge locations to ensure swift and efficient content delivery, enhancing user experiences across the digital landscape.

3. Edge Location : Optimizing Content Delivery

Edge locations are sites used by AWS for content delivery and caching.

Used mainly by:

- CloudFront
- Route 53

Purpose:

- Reduce latency
- Faster content delivery